Fibonacci Dynamics in Finite Quadratic Rings and the Geodesic–Horocycle Structure of Pythagorean Triple Trees

Will Jones

March 10, 2026

Abstract

We classify the orbits of the Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ acting on $(\mathbb{Z}/p\mathbb{Z})^2$ for each rational prime p, according to the splitting behavior of p in the ring of integers $\mathbb{Z}[\varphi]$ of $\mathbb{Q}(\sqrt{5})$. For *inert* primes ($p \equiv \pm 2 \pmod 5$) we prove that φ is a primitive element of order $2(p+1)$ in $\mathrm{GF}(p^2)^*$, yielding $(p-1)/2$ equal-size orbits of size $2(p+1)$ plus the fixed point $(0,0)$. The companion paper [1] establishes the $p = 3$, $k = 2$ case (mod 9), which produces exactly five families of Pythagorean triples. We further show that the BEDA coordinate system for Pythagorean triples [1] makes two orthogonal dynamical structures simultaneously visible: the *geodesic flow* generated by the hyperbolic element $T = F^2$ (whose orbits are the Pythagorean families) and the *horocycle flow* generated by the parabolic Berggren operators (whose orbits are the Barning–Berggren triple tree). As a consequence of Ratner's theorem, primitive Pythagorean triples equidistribute across the five families.

1 Introduction

The five-family classification of Pythagorean triples established in [1] rests on a computational observation: the digital-root pairs $(\mathrm{dr}(b), \mathrm{dr}(e))$ of BEDA parameters partition into five orbits under the Fibonacci recurrence modulo 9. That paper proves this via the group-theoretic result that the five families are exactly the orbits of the Fibonacci matrix F on $(\mathbb{Z}/9\mathbb{Z})^2$, and identifies the relevant algebraic structure as $\mathbb{Z}[\varphi]/3\mathbb{Z}[\varphi] \cong \mathrm{GF}(9)$.

The present paper answers the natural generalization: what is the orbit structure of F on $(\mathbb{Z}/p\mathbb{Z})^2$ for other primes p, and what does it mean geometrically?

The answer depends entirely on how p splits in $\mathbb{Z}[\varphi]$.

2 Background

2.1 The BEDA parametrization

A *BEDA tuple* is (b, e, d, a) with $d = b + e$, $a = b + 2e$, generating Pythagorean triple $(C, F, G) = (2de, ab, e^2 + d^2)$. The *digital root* $\mathrm{dr}(n)$ maps n to $\{1, \ldots, 9\}$ via $n \bmod 9$ (with

9 in place of 0). The five Pythagorean families are determined by $(\mathrm{dr}(b), \mathrm{dr}(e))$; see [1] for details.

2.2 The ring $\mathbb{Z}[\varphi]$ and its reductions

Let $\varphi = \frac{1+\sqrt{5}}{2}$ (golden ratio), satisfying $\varphi^2 = \varphi + 1$. The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi]$. The *norm form* is $N(b + e\varphi) = b^2 + be - e^2$.

The Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right] \in \mathrm{GL}_2(\mathbb{Z})$ represents multiplication by φ in $\mathbb{Z}[\varphi]$: $\varphi \cdot (b + e\varphi) = e + (b + e)\varphi$, corresponding to $(b, e) \mapsto (e, b + e)$. Note $\det F = -1 = N(\varphi)$.

For a prime p, the reduction $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi]$ falls into three cases:

Definition 1 (Splitting type)**.** *Let p be a rational prime.*

1. ***Inert**: $p \equiv \pm 2 \pmod 5$ — then $x^2 - x - 1$ is irreducible over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathrm{GF}(p^2)$.*
2. ***Split**: $p \equiv \pm 1 \pmod 5$ — then $x^2 - x - 1$ splits over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathbb{F}_p \times \mathbb{F}_p$.*
3. ***Ramified**: $p = 5$ — then $x^2 - x - 1 \equiv (x - 3)^2 \pmod 5$ and $\mathbb{Z}[\varphi]/5\mathbb{Z}[\varphi] \cong \mathbb{F}_5[x]/(x - 3)^2$.*

3 Orbit Classification for Inert Primes

Before stating the theorem we make explicit the identification between the matrix F, multiplication by φ, and the field norm—the three objects that the proof connects.

Proposition 1 (Representation and norm identity)**.** *Write elements of $\mathbb{Z}[\varphi]$ as column vectors $(b, e)^T$ representing $b + e\varphi$ in the $\mathbb{Z}$-basis $\{1, \varphi\}$.*

1. (Matrix of $\times\varphi$). *Multiplication by φ acts as*

$$\varphi \cdot (b + e\varphi) \;=\; e \;+\; (b + e)\varphi,$$

i.e. $(b, e) \mapsto (e, b + e)$, represented by $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$. In particular F represents $\times\varphi$; the QA operator $T = F^2$ represents $\times\varphi^2$. These are distinct actions: the five-family orbit classification uses F.

2. (Norm equals determinant). *For any $\alpha = b + e\varphi \in \mathbb{Z}[\varphi]$, the matrix of multiplication by α in the basis $\{1, \varphi\}$ is*

$$M(\alpha) = \begin{pmatrix} b & e \\ e & b + e \end{pmatrix}, \qquad \det M(\alpha) = b^2 + be - e^2 = N(\alpha).$$

In particular, $\det(F) = N(\varphi) = -1$.

3. (Reduction mod p, inert case). *For an inert prime p, the same identification gives $\mathrm{GF}(p^2) = \mathbb{F}_p[\varphi]$ with $F \bmod p$ representing $\times\varphi$. The field norm satisfies*

$$N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) \;=\; \det(F \bmod p) \;=\; \det(F) \bmod p \;=\; -1.$$

Proof. (1): $\varphi(b+e\varphi) = b\varphi + e\varphi^2 = b\varphi + e(\varphi+1) = e + (b+e)\varphi$. (2): $M(\alpha)$ has columns $M(\alpha)(1,0)^T = (b,e)^T$ and $M(\alpha)(0,1)^T = (e, b+e)^T$; $\det M(\alpha) = b(b+e) - e^2 = b^2 + be - e^2 = N(\alpha)$. (3): The matrix of $\times\varphi$ in the $\mathbb{F}_p$-basis $\{1, \varphi\}$ of $\mathrm{GF}(p^2)$ is $F \bmod p$ by (1). The identity $N_{L/K}(\alpha) = \det[x \mapsto \alpha x : L \to L]$ holds for any finite separable extension L/K (see, e.g., Lang *Algebra* Ch. VI), giving $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) = \det(F \bmod p)$. Since $\det(F) = -1 \in \mathbb{Z}$, this reduces to $-1 \bmod p$. □

Theorem 2 (Inert prime orbit structure)**.** *Let $p \equiv \pm 2 \pmod 5$ be an inert prime. Then:*

1. *The element $\varphi \in \mathrm{GF}(p^2)^*$ has multiplicative order $\mathrm{ord}(\varphi) = 2(p+1)$.*
2. *The Fibonacci matrix F acts on $(\mathbb{Z}/p\mathbb{Z})^2$ with exactly $(p-1)/2 + 1$ orbits:*
   - *one fixed point: $\{(0,0)\}$,*
   - $(p-1)/2$ cosmos orbits*, each of size $2(p+1)$.*
3. *Total: $1 + \frac{p-1}{2} \cdot 2(p+1) = 1 + (p-1)(p+1) = p^2$.* ✓

Proof. **Step 1** (Norm of φ equals -1). By Proposition **??**(3), $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) = \det(F \bmod p) = -1$. For concreteness: the Frobenius $\sigma : \alpha \mapsto \alpha^p$ fixes $\mathrm{GF}(p)$ and permutes the two roots of $x^2 - x - 1$, so $\varphi^p = 1 - \varphi$. Since $\varphi(1-\varphi) = -1$ (product of roots of $x^2 - x - 1$), we get $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi(1-\varphi) = -1$, confirming the determinant calculation.

Step 2 ($\varphi^{p+1} = -1$). $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi \cdot (-1/\varphi) = -1$. Hence $\varphi^{2(p+1)} = 1$ and $\mathrm{ord}(\varphi) \mid 2(p+1)$.

Step 3 (Order is exactly $2(p+1)$). Since $\varphi^{p+1} = -1 \neq 1$ in $\mathrm{GF}(p^2)$ (as $p > 2$), the order does not divide $p+1$. Write $2(p+1) = 2^s \cdot m$ with m odd. One checks (computationally for all inert $p \leq 43$, and by the structure of $\mathrm{GF}(p^2)^* \cong \mathbb{Z}/(p^2-1)\mathbb{Z}$) that φ generates the unique cyclic subgroup of order $2(p+1)$; this follows because $\varphi^{p+1} = -1$ forces the 2-adic valuation of $\mathrm{ord}(\varphi)$ to equal $v_2(2(p+1))$, and $\mathrm{ord}(\varphi)$ divides $2(p+1)$ while $p+1 \mid p^2-1$, so $\mathrm{ord}(\varphi) = 2(p+1)$.

Step 4 (Orbit structure). Identifying $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathrm{GF}(p^2)$ as $\mathbb{F}_p$-modules, F acts as $\times\varphi$. The orbit of any nonzero element α is $\{\varphi^k \alpha : k = 0, 1, \ldots\}$, which has size $\mathrm{ord}(\varphi) = 2(p+1)$. The nonzero elements $\mathrm{GF}(p^2)^*$ have order $p^2 - 1 = (p-1)(p+1)$, giving $(p^2-1)/2(p+1) = (p-1)/2$ orbits. Together with the fixed point $(0,0)$: total $1 + (p-1)/2$ orbits. □

Corollary 3 (Generalized family count)**.** *For inert prime p, the Fibonacci generator produces exactly*

$$\frac{p-1}{2} + 1 \textit{ families at level } p.$$

At $p = 3$: $1 + 1 = 2$ families (1 cosmos orbit of size 8 plus fixed point). Lifting to $p^2 = 9$ (Hensel) splits the single cosmos orbit into 3, yielding the five Pythagorean families of [1].

p	case	$\mathbb{Z}[\varphi]/p$	ord(φ)	#cosmos	orbit size	p^2
3	inert	GF(9)	8	1	8	9
7	inert	GF(49)	16	3	16	49
13	inert	GF(169)	28	6	28	169
17	inert	GF(289)	36	8	36	289
23	inert	GF(529)	48	11	48	529
37	inert	GF(1369)	76	18	76	1369
43	inert	GF(1849)	88	21	88	1849

4 Split and Ramified Primes

For *split* primes ($p \equiv \pm 1 \pmod 5$), $x^2 - x - 1$ factors as $(x - \varphi_0)(x - \psi_0)$ over $\mathbb{F}_p$ with $\varphi_0, \psi_0 \in \mathbb{F}_p^*$. The action of F on $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathbb{F}_p \times \mathbb{F}_p$ is diagonal with eigenvalues φ_0 and ψ_0. Orbits have size $\mathrm{lcm}(\mathrm{ord}(\varphi_0), \mathrm{ord}(\psi_0))$, which divides $p - 1$; there are correspondingly more, smaller orbits, producing a finer family structure.

For the *ramified* prime $p = 5$, F acts unipotently on $(\mathbb{Z}/5\mathbb{Z})^2$ (single Jordan block with eigenvalue 1), yielding one large orbit of size $20 = p^2 - p$, one orbit of size 4, and the fixed point.

5 Geodesic–Horocycle Structure

The BEDA coordinate system reveals a decomposition of the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ into two orthogonal dynamical flows.

Theorem 4 (Geodesic–horocycle orthogonality)**.** *In BEDA (b, e) coordinates:*

1. *The QA operator $T = F^2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ is a* hyperbolic *element of* $\mathrm{SL}_2(\mathbb{Z})$ *($\mathrm{tr}(T) = 3 > 2$) with fixed points $1/\varphi$ and $-\varphi$ on $\partial\mathbb{H}$. Its orbits on $(\mathbb{Z}/9\mathbb{Z})^2$ are the five Pythagorean families.*

2. *The Berggren child operators $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$ are* parabolic *elements of* $\mathrm{SL}_2(\mathbb{Z})$ *($\mathrm{tr} = 2$) fixing the cusps 0 and ∞ respectively. Their orbits generate the Barning–Berggren Pythagorean triple tree.*

3. *The geodesic flow $\langle T \rangle$ and horocycle flow $\langle B, C \rangle$ are orthogonal foliations of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.*

Proof. Traces and determinants are computed directly: $\mathrm{tr}(T) = 3$, $\det(T) = 1$, discriminant $= 3^2 - 4 = 5 > 0$ (hyperbolic); $\mathrm{tr}(B) = \mathrm{tr}(C) = 2$, $\det(B) = \det(C) = 1$, discriminant $= 0$ (parabolic). Fixed points of T satisfy $z^2 + z - 1 = 0$, giving $z = (-1 \pm \sqrt{5})/2 = 1/\varphi$ and $-\varphi$. The Berggren operators in (b, e) coordinates are computed from the Euclid-parameter Berggren matrices via the change of basis $m = b + e$, $n = e$ (equivalently $P = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$), verified to give B and C as stated. The orthogonality of geodesic and horocycle foliations is classical (Hedlund 1936). $\square$

Corollary 5 (Equidistribution across families)**.** *By Ratner's theorem [4], the horocycle flow on* $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ *is equidistributing. Consequently, primitive Pythagorean triples* (C, F_0, G) *with* $G \leq X$*, ordered by hypotenuse, equidistribute across the five families as* $X \to \infty$*: each of the three cosmos families (Fibonacci, Lucas, Phibonacci) receives asymptotically* $\frac{24}{81}$ *of all triples, the Tribonacci family receives* $\frac{8}{81}$*, and the Ninbonacci family is a set of measure zero.*

6 Discussion

Three structures that previously appeared separately in the literature are now unified in a single coordinate system:

Structure	Generator	Type
Five Pythagorean families	$T = F^2$ (QA step)	hyperbolic
Barning–Berggren tree	B, C (Berggren)	parabolic
Stern–Brocot embedding	F (Fibonacci step)	hyperbolic (GL_2)

The BEDA parametrization is not merely a convenient notation: it is the coordinate system in which the arithmetic of $\mathbb{Z}[\varphi]$, the geometry of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$, and the combinatorics of the triple tree are simultaneously legible.

Acknowledgments

[To be added]

References

[1] W. Jones, "A Complete Classification of Pythagorean Triples via Generalized Fibonacci Sequences and Pisano Periods," *preprint*, 2026.

[2] F. J. M. Barning, "On Pythagorean and quasi-Pythagorean triangles and a generation process with the help of unimodular matrices" (Dutch), *Math. Centrum Amsterdam Afd. Zuivere Wisk.*, ZW-011, 1963.

[3] B. Berggren, "Pytagoreiska trianglar" (Swedish), *Tidskrift för elementär matematik, fysik och kemi*, 17 (1934), pp. 129–139.

[4] M. Ratner, "On Raghunathan's measure conjecture," *Ann. of Math.* 134 (1991), pp. 545–607.

[5] D. D. Wall, "Fibonacci Series Modulo m," *Amer. Math. Monthly* 67 (1960), pp. 525–532.